

Question Set 7

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions

- 1) Have a look at the initiator structures in Lecture Notes 3 and predict which ones can be used for suspension polymerisation and which ones for emulsion polymerisation.

If the initiator is water soluble: emulsion (AIBA, hydrogen peroxide)

If the initiator is hydrophobic/soluble in organic solvents: suspension. (all the peroxides we've seen in LN3, with the exception of hydrogen peroxide, AIBN)

- 2) Have a look at the monomer structure in Lecture Notes 3, and also in Question Set 6, and predict if any monomers may be difficult to polymerise through suspension or emulsion polymerisation. How would you solve this problem?

All the monomers soluble in water (for example, acrylic acid and methacrylic acid). You can do an inverse emulsion or inverse suspension. For emulsion: you will have to use an initiator soluble in organic solvents.

- 3) Please do not use the internet or a book for this question: (the main point is NOT to get the question right, but understand the main principles behind it) (For all the cases, please answer WHY you have proposed such structure):

Design 3 molecules that you have not seen in the Lecture Notes that can potentially be a surfactant (the three should be classified differently from each other). Classify them according to surfactant type.

Any molecule that has one part soluble in water and another very hydrophobic: see LN4.

- 4) In an experiment styrene is polymerised under three different heterogeneous conditions. In all those, styrene is appropriately dispersed in water containing a suitable initiating species and heated to 60 °C. The three different initiators (AIBA, AIBN, PPS) are shown below, together with the corresponding decomposition rate constants (k_d) at 60°C in a suitable solvent/medium in which they are soluble.

PLEASE UNDERSTAND: any initiator can be used in suspension or emulsion polymerization. The fact we're stating below that, for example, AIBN is suspension, is because the monomer is styrene (hydrophobic), therefore, if initiator is hydrophobic (AIBN) a heterogeneous system will be suspension, and the opposite would be emulsion. BUT, if the monomer was water soluble, AIBN would be emulsion, and the other two suspension!

- a) Indicate what sort of polymerisation type corresponds to the use of AIBA, AIBN and PPS, respectively. Explain the basis for your answer

This depends on whether the initiator is soluble in water or not. AIBA and PPS are soluble in water. Therefore, as the styrene is hydrophobic, this means the heterogeneous system will be emulsion. AIBN is hydrophobic, therefore, will be placed with the monomer: suspension.

- b) Indicate the most suitable conditions among the three to synthesise the highest MW polystyrene. For your answer, assume that [styrene] (10mM) and [initiator] (10μM) are the same in all cases, that dispersion has worked equally well in all cases, and that the same number of dispersed polymerization nuclei have been formed. Assume the three initiators have the same efficiency of initiation with respect to styrene at the working temperature ($f \sim 0.6$). Hint: you can calculate ratios of DP with regards to each other. (Note: it is

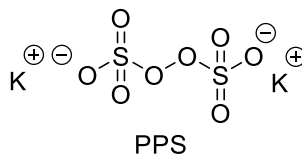
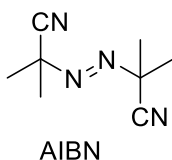
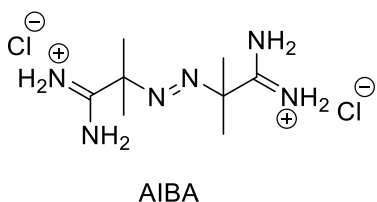
acceptable to express this ratio as a function of unknown constants or parameters and discuss thereof). Use suitable equations in your answer explaining your choice for such equations.

You should already know, before doing any calculations that the suspension system (kinetics as per LN3, normal FRP kinetics) would provide lower MW than emulsion (special kinetics as per LN4). ALSO, that if you are comparing two initiators of the same type (both suspension or both emulsion) then, the initiator with the highest k_d (more FR) will lead to lower MW (because termination is second order with respect to $[M^*]$, while propagation and initiation is first order). SO, without doing any calculation, clearly emulsion (AIBA and PPS) higher MW than suspension (AIBN) / (see also equations where k_d is part of denominator. Then, between AIBA and PPS, the one with lower k_d , will have higher MW (also k_d in denominator in equation). SO: MW AIBA > PPS > AIBN.

You can also work this out using equations. The equations for kinetics (suspension, AIBN, LN3; emulsion, AIBA & PPS, LN3). If you calculate ratios of average kinetic chain length, you need to divide any by any other of the initiators. Thus the $[\text{styrene}]$ and initiators and f will cancel out. The emulsion kinetics will depend on N (number of micelles, which we don't know: BUT we DO know that N is a VERY large number).

- c) Explain why two of the systems are markedly superior to the third in terms of the MW values that can be obtained.

The reason is that two (AIBA and PPS) are emulsion: this means the FR initiator is not in contact with the monomer directly. Therefore, this reduces the $[M^*]$ in presence with the propagating FR. Because termination is second order with respect to $[M^*]$, this means termination will be a lot more reduced in rate than propagation and initiation. The result, is a higher MW. In suspension, the FR initiator is in contact with the monomer, so, much higher $[M^*]$, more termination, lower MW.



Azo-bis(isobutyramidine hydrochloride) AIBA: k_d (60 °C, suitable solvent): $3.70 \times 10^{-5} \text{ s}^{-1}$

Azo-bis(isobutyronitrile) AIBN: k_d (60 °C, suitable solvent): $9.80 \times 10^{-6} \text{ s}^{-1}$

Potassium persulphate: k_d (60 °C, suitable solvent): $1.08 \times 10^{-1} \text{ s}^{-1}$

Data for styrene: Styrene (60 °C): $k_p = 176 \text{ L mol}^{-1} \text{ s}^{-1}$; $k_t = 72 \times 10^6 \text{ L mol}^{-1} \text{ s}^{-1}$

Practice questions (only if there is time, and revision at the end of module):

- 5) Explain why in NAD, polymer particles formed are smaller than in Precipitation Polymerisation.
Because in NAD there is a polymer system that stabilises the growing polymer. Smaller particles have a much larger surface area proportion to their volume, compare to larger particles. This exposed surface area is unstable (the polymer precipitates because it is not soluble in the chosen solvent). However, a stabiliser will provide a more stable interface polymer-solvent, which can grow larger (particles smaller).
- 6) Propose different media polymerisations for the following applications, and explain why, as well as advantaged and disadvantages:
 - a) A very high MW polymer **Emulsion**
 - b) A oil-based paint (oil, in this case, means hydrophobic liquid medium) **–Solution, organic solvent**
 - c) A water-based paint **–Emulsion (very typical in water-based paints) or occasionally solution in water (unlikely)**

- d) A block of polymer –this is NOT a block copolymer. Could be bulk.
- e) Spherical beads of μm size Suspension
- f) Spherical beads of nm size Emulsion
- g) Beads for grinding/extruding and other polymer processing suspension
- h) Threads and fibers Interface polymerisation.

7) Draw the mechanism of the polymerisation leading to Nylon 4,4

See LN2 and videos for any amide.

8) Draw the mechanism of the suspension polymerisation of styrene and divinyl benzene (propose a suitable initiator). For full marks, always show initiator degradation, all resonance forms for resonance-stabilised reactive species, mechanisms of initiation, propagation and all termination pathways. Draw the structure of the polymer. SEE LN3 and videos.